

Feature scaling

Standardization
(previous Note)

Normalization
(this Note)

Min-max scaling

$$\frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

range 0 to 1

use: known min, max

Mean normalization

$$\frac{x - \bar{x}}{x_{\max} - x_{\min}}$$

range -1 to 1

use: centred data

Max-abs scaling

$$\frac{x}{|x|_{\max}}$$

range -1 to 1

use: sparse data

Robust scaling

$$\frac{x - \text{median}}{\text{IQR}}$$

no fixed range

use: outliers